

Mate Six Connector Types SOP (1x Episode: 6 Connectors)

One episode mates all six connectors on one connector panel. The table begins with the **connector panel** standing on the **assembly spot**, just right of the center of the table. Six **ports** sit in a row across the panel, all facing **up**, numbered **1** at the left to **6** at the right: port 1 an **RJ45 jack**, port 2 a **USB port**, port 3 a **barrel jack**, port 4 a **header shroud**, port 5 a **spade tab**, port 6 a **ring stud** with its **thumb nut** beside it. A **lead tray** holding six **leads** stands in the **start zone**, one lead per numbered **tray slot**, each lead ending in the connector for the port with the same number.

This is a single-arm task. Only the **right arm** works. The **left arm stays at home with its gripper open for the whole episode** and touches nothing. Because there is no second gripper to hold or steady anything, the fixtures do that job: the connector panel is a heavy plate with a non-slip underside, so a connector can be pushed into a port and the thumb nut turned without the panel sliding, and every port faces up so each push goes **straight down** into the table instead of shoving the panel sideways. Nothing is ever held in the air while something else is done to it.

Every connector that has a correct way round is mated with its keyed feature facing the **front edge** — the RJ45 latch tab, the USB symbol, and the header key ridge all point the same way. The barrel plug is round and has no correct way round; for it, straight and in line is the whole job.

The order never changes: mate all six connectors, working port 1 through 6; then tug-test all six, again 1 through 6. No connector is tug-tested until all six are mated. The episode does not end until every one of the six connectors is fully mated and has held its tug test.

The table is set up in one of three ways. Only the lead tray moves; the connector panel, the assembly spot, and the nut dish are in the same place in all three.

- **Config M:** the lead tray is at the front-center, in front of the assembly spot, no part of it left of the table's center line.
- **Config R1:** the lead tray is at the front-right, right of the assembly spot.
- **Config R2:** the lead tray is at the back-right, behind the panel and the nut dish.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the lead tray starts at the front-center (**Config M**), the front-right (**Config R1**) or the back-right (**Config R2**). One config per episode, chosen before recording and

never changed mid-episode.

- **Same-side rule:** this is a single-arm task, so the **right gripper** picks every lead in every config; the config changes only the direction of the reach to the tray. The left side of the table is not used, because the left arm is parked and the right arm never leans across the table's center line, so no left zone is reachable. Nothing is handed over.
- **Fixed roles:** everything else is the same in all three configs — the panel stays on the assembly spot, the thumb nut starts in the nut dish, the left arm stays at home, and the leads go in port 1 through 6.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the connector panel with all six ports, the nut dish beside port 6, the lead tray in the start zone for this episode's config, and both arms.
3. The connector panel is visible from above and from the front, so all six ports, the face of each port, the thumb nut, and the connector on each lead can be seen.
4. Both arms are at home with grippers open.
5. The table is bare apart from the connector panel, the nut dish, the lead tray, and robot hardware.
6. The right arm reaches every tray slot in the start zone for this episode's config, every port from 1 to 6, and the nut dish without stretching and without leaning across the table's center line.
7. The left arm is at home, gripper open, and stays there.

Materials checklist

1. One **connector panel** stands on the **assembly spot**, a taped outline just right of the center of the table, **square** to the front edge and flat on the table.
2. The panel is a **heavy plate with a non-slip underside**, so a connector can be pushed straight down into a port and the thumb nut can be turned without the panel sliding or lifting. It is never gripped.

3. The panel is narrow enough that port 1, at its left end, still sits right of the table's center line, so the right arm reaches the whole row without leaning across the table.
4. Six **ports** are fixed in a row across the panel, all facing **up**, numbered **1** at the left to **6** at the right. The numbers are printed on the panel beside each port and readable from above.
5. The ports are, left to right: port 1 **RJ45 jack**, port 2 **USB port**, port 3 **barrel jack**, port 4 **header shroud**, port 5 **spade tab**, port 6 **ring stud**.
6. Port 1's RJ45 jack has its **latch slot** on the front wall. Port 2's USB port has its **plastic tongue** at the back of the opening. Port 4's header shroud has its **notch** in the front wall. Port 5's spade tab stands up from the panel with its flat faces toward the front and the back. Port 6's ring stud stands up bare, with its **thumb nut** resting in the **nut dish** just right of the stud.
7. All six ports are empty, clean, and undamaged. No port holds a connector at the start.
8. The **lead tray** stands in the start zone for this episode's config, the other two zones bare, with six **tray slots**, numbered 1 at the left to 6 at the right. Each slot is labeled with its number and its connector name, and holds that one lead: slot 1 RJ45, slot 2 USB, slot 3 barrel, slot 4 header, slot 5 spade, slot 6 ring.
 - **Config M:** front-center, in front of the assembly spot, no part of it left of the center line
 - **Config R1:** front-right, right of the assembly spot
 - **Config R2:** back-right, behind the panel and the nut dish
9. The lead tray is heavy and non-slip, so a lead can be lifted straight out of a slot without the tray sliding or tipping.
10. All six **leads** are cut to the same length. Each has one connector at one end and a plain cut end at the other. Each connector is straight on its lead, with no bent pin, no bent blade, and no cracked shell.
11. The **thumb nut** for port 6 is knurled, turns by hand, and starts in the **nut dish**, not on the stud. The nut dish is weighted so the nut can be picked out of it without the dish moving.
12. Before collection, confirm by hand that:
 - each connector goes into its own port straight down, without force;
 - a connector can be pushed fully home without the panel sliding or lifting;
 - the RJ45 plug clicks in its jack and the click can be heard and seen;
 - the USB plug goes fully in with its shell flush, and will not go in the wrong way round;
 - the barrel plug slides on and off the barrel jack without binding;
 - the header plug drops into the shroud one way only, and stops if the key ridge is turned the wrong way;
 - the spade receptacle slides onto the tab with a firm, even grip;
 - the thumb nut runs down the stud by hand and stops solid, without the panel turning under it;

- a mated connector holds a light pull.

Workspace layout

Everything below is a fixed area of the table, judged by eye against the table edges, the taped outline, and the fixtures standing on it. The three areas sit in a short arc, all inside the right arm's reach.

- **Just right of center — assembly spot:** one taped outline. The connector panel stands here and does not move.
- **Across the panel:** ports 1 to 6, left to right, all facing up.
- **Right of port 6 — nut dish:** the thumb nut for the ring stud.
- **Start zone — lead tray:** six numbered tray slots holding the six leads, at the front-center (**Config M**), the front-right (**Config R1**), or the back-right (**Config R2**).

Arm assignment

- **Right gripper — all of the work:** picks each lead from the tray, turns each connector the right way round, pushes it straight down into its port, drops the ring terminal over the stud, turns the thumb nut down, and tug-tests every lead. It reaches forward and to the left for the tray in Config M, forward and to the right in Config R1, and back past the nut dish in Config R2.
- **Left arm — parked:** stays at home with its gripper open from the start of recording to the end, in every config. It holds nothing, steadies nothing, and reaches for nothing.

Vocabulary

- **Assembly spot:** the taped outline just right of the center of the table where the connector panel stands. The panel stays inside it for the whole episode.
- **Square:** the front edge of the panel lines up with the front edge of the table, so the row of ports runs level across the panel and neither end of the panel sits nearer the front.
- **Connector panel:** the heavy plate the six ports are mounted on. Its underside is non-slip, so it holds itself still while a connector is pushed in. It is never gripped and never lifted.
- **Port:** one mating place on the panel. Each one has its own number and takes one kind of connector.
- **Port number:** the number printed on the panel beside each port, 1 at the left to 6 at the right.
- **Facing up:** the opening or the stud of the port points at the ceiling, so a connector comes into it straight down from above.
- **Front edge:** the edge of the table nearest the collector. Keyed connectors are mated with their keyed feature pointing this way.
- **Lead:** one cable with a connector on one end. Six leads are mated in one episode.

- **Lead tray:** the heavy holder in the start zone with six numbered slots, one lead per slot.
- **Start zone:** where the lead tray stands at the start of the episode — front-center (**Config M**), front-right (**Config R1**), or back-right (**Config R2**). One per episode, chosen before recording and never changed mid-episode.
- **Tray slot:** one numbered slot in the lead tray. The lead in slot 3 goes in port 3, and so on.
- **Connector:** the fitting on the end of a lead — the part that goes into the port.
- **Keyed feature:** the one part of a connector that says which way round it goes: the RJ45 latch tab, the USB symbol, or the header key ridge.
- **Straight down:** the connector comes down in line with its port, with the lead standing up above it, and does not lean, tilt, or come in from the side.
- **Cocked:** the connector is leaning in its port instead of sitting in line with it. A cocked connector jams part way in.
- **Fully mated:** the connector has gone as far into its port as it goes, shown by the tell for that port given in its substep, and it stays there when the right gripper opens.
- **Half-mated:** the connector has started into its port but has not reached its tell — a gap at the face, no click, or a shell still standing proud.
- **RJ45 plug:** the eight-contact plug on lead 1, with a springy **latch tab** on one face.
- **Latch tab:** the springy plastic tongue on the RJ45 plug. It faces the front edge and drops into the latch slot when the plug is fully mated.
- **Click:** the small snap heard and seen when the RJ45 latch tab drops into the latch slot. Until it clicks, the plug is not fully mated.
- **USB plug:** the flat metal-shelled plug on lead 2, with the **USB symbol** stamped on one face.
- **USB symbol:** the small trident mark on one face of the USB plug. It faces the front edge.
- **Shell:** the metal sleeve around the USB plug.
- **Flush:** the metal shell has gone in far enough that the plastic body of the plug sits down on the face of the port with no gap showing.
- **Barrel plug:** the round plug on lead 3, with no correct way round.
- **Sleeve face:** the flat step on the barrel plug where the round body ends. It touches the face of the barrel jack when the plug is fully mated.
- **Header plug:** the plastic housing on lead 4, with a raised **key ridge** along one face.
- **Key ridge:** the raised rib on one face of the header plug. It faces the front edge and lines up with the notch in the shroud.
- **Shroud:** the plastic wall around the header pins on port 4, with a notch in its front wall.
- **Bottomed:** the header housing has gone all the way into the shroud and its rim sits down on the top of the shroud all the way round.
- **Spade receptacle:** the flat metal sleeve on lead 5 that slides onto the spade tab.

- **Spade tab:** the flat metal blade standing up at port 5, with its flat faces toward the front and the back.
- **Flats lined up:** the two flat faces of the spade receptacle lie along the two flat faces of the tab, so the receptacle slides on straight instead of catching the tab's edge.
- **Collar:** the plastic sleeve at the back of the spade receptacle where the lead enters it.
- **Ring terminal:** the flat metal ring on lead 6 that drops over the ring stud.
- **Ring stud:** the threaded post standing up at port 6.
- **Lying flat:** the ring terminal sits down on the panel all the way round the stud, with no part of it lifted off the panel.
- **Thumb nut:** the knurled nut that turns down the ring stud onto the ring terminal.
- **Nut dish:** the weighted dish just right of port 6 that the thumb nut starts in and returns to at reset.
- **Nut closed:** the thumb nut has been turned clockwise until it stops turning and the ring terminal no longer moves under it.
- **Tug test:** a light, steady pull on a lead, straight up away from the panel, for a count of one.
- **Holds:** the connector does not move at all during its tug test.
- **Backs off:** the connector lifts, slides, or comes out of its port during its tug test.
- **Lying clear:** a mated lead rests on the panel or the table without lying over another port.

Steps

Run Steps 1 and 2 in order on the one connector panel, then end the episode with Step 3.

Only Step 1 depends on where the lead tray is: the direction of the reach to the tray and of the carry to the port, set out once on the IF lines at the top of Step 1 and the same for all six substeps. Every other line in Steps 1 to 3 is the same in all three configs.

Through all three steps the **left arm** stays at home with its gripper open. The **connector panel** is never gripped, held, or steadied — its weight and its non-slip underside do that.

Step 1: Mate all six connectors, port 1 through 6

Goal: all six connectors are fully mated in their own ports, each one the right way round.

Look where the lead tray is before reaching for the first lead.

- **IF the lead tray is at the front-center (Config M):** the **right gripper** reaches forward and to the left, in front of the assembly spot, for each lead, lifts it straight up, and carries it back and to the right to its port.
- **IF the lead tray is at the front-right (Config R1):** the **right gripper** reaches forward, right of the assembly spot, for each lead, lifts it straight up, and carries it back and to the left to its port.

- **IF the lead tray is at the back-right (Config R2):** the **right gripper** reaches back, past the nut dish, for each lead, lifts it straight up high enough to clear the nut dish, the panel, and the mated leads, and carries it forward to its port.

Then, in all three:

- Work **port 1 first, then 2, 3, 4, 5, 6**. Do one lead at a time.
- The lead in **tray slot N** goes in **port N**. Take the lead from the slot with the same number as the port being mated.
- With the **right gripper**, carry each lead in one go, straight from the tray to its port. Do not set a lead down on the panel or the table on the way.
- Hold each lead by its **connector** with the **right gripper**, across the body of the connector. Never pinch the pins, the blade, or the metal shell.
- Every connector goes in **straight down**, so the push presses the panel onto the table instead of pushing it sideways. The **right gripper** never levers a connector sideways and never forces one that has stopped.
- If a connector will not go in, lift it clear with the **right gripper**, turn it the right way round, line it up again, and push it straight down again.
- After each connector, lay its lead down **lying clear** of the other ports with the **right gripper**.

1.1 Port 1 — RJ45 plug

- With the **right gripper**, pick the lead from **tray slot 1** and lift it straight up.
- Turn the plug in the **right gripper** so its **latch tab** faces the **front edge**.
- With the **right gripper**, put the plug in the mouth of the jack and push it **straight down** until it **clicks**.
- Open the **right gripper** and let go.
- **Check:** the click was heard and seen, the latch tab has dropped into the latch slot, and the plug does not stand proud of the jack. If there was no click, lift the plug clear with the **right gripper** and push it straight down again.

1.2 Port 2 — USB plug

- With the **right gripper**, pick the lead from **tray slot 2** and lift it straight up.
- Turn the plug in the **right gripper** so the **USB symbol** faces the **front edge**.
- With the **right gripper**, put the plug in the mouth of the port and push it **straight down** until the **shell** is **flush**.
- Open the **right gripper** and let go.
- **Check:** the shell is flush and no gap shows between the plug body and the face of the port. If the plug stops part way, lift it clear with the **right gripper**, turn it so the symbol faces the front edge, and push it straight down again. Never force a USB plug that has stopped — it is the wrong way round.

1.3 Port 3 — barrel plug

- With the **right gripper**, pick the lead from **tray slot 3** and lift it straight up.
- The barrel plug is round and has no correct way round. Line it up **in line with the jack** in the **right gripper**, lead standing straight up.
- With the **right gripper**, push it **straight down** until the **sleeve face** touches the face of the jack.
- Open the **right gripper** and let go.
- **Check:** the plug stands in line with the jack, not **cocked**, and no gap shows at the sleeve face. If the plug is cocked or a gap shows, lift it clear with the **right gripper**, stand it in line, and push it straight down again.

1.4 Port 4 — header plug

- With the **right gripper**, pick the lead from **tray slot 4** and lift it straight up.
- Turn the housing in the **right gripper** so its **key ridge** faces the **front edge**, lined up with the **notch** in the front wall of the **shroud**.
- With the **right gripper**, lower the housing into the shroud and push it **straight down** until it is **bottomed**.
- Open the **right gripper** and let go.
- **Check:** the housing rim sits down on the top of the shroud all the way round, and no pin is bent. If the housing stops high, lift it clear with the **right gripper**, turn the key ridge to the front edge, and push it straight down again. Never force a header plug that has stopped high.

1.5 Port 5 — spade receptacle

- With the **right gripper**, pick the lead from **tray slot 5** and lift it straight up.
- Turn the receptacle in the **right gripper** so its **flats** are **lined up** with the flat faces of the **spade tab**, and its **collar** points up.
- With the **right gripper**, bring the mouth of the receptacle over the top of the tab and push it **straight down** until the **collar** meets the base of the tab.
- Open the **right gripper** and let go.
- **Check:** the receptacle sits square on the tab, the collar is down at the base of the tab, and the tab is not bent over. If the receptacle catches the edge of the tab, lift it clear with the **right gripper**, line the flats up, and push it straight down again.

1.6 Port 6 — ring terminal and thumb nut

- With the **right gripper**, pick the lead from **tray slot 6** and lift it straight up.
- With the **right gripper**, drop the **ring terminal** over the **ring stud** and press it down until it is **lying flat** on the panel, with the lead running toward the **front edge**.
- Open the **right gripper** and let go. The ring stays lying flat on its own.

- With the **right gripper**, pick the **thumb nut** out of the **nut dish** and start it on the stud, turning **clockwise**.
- With the **right gripper**, turn the nut **clockwise** a quarter turn, lift clear, re-grip, and turn another quarter turn. Repeat until the nut stops turning.
- Never turn more than a quarter turn in one grip, never turn the nut counter-clockwise, and never keep forcing it once it has stopped.
- **Check:** the ring terminal is lying flat, the lead runs toward the front edge, and the **nut is closed** with the ring no longer moving under it. If the ring lifted while the nut turned, open the nut in quarter turns with the **right gripper**, press the ring flat again, and close the nut again.

Check: all six ports hold their own connector, each is **fully mated** by the tell in its substep, and no lead lies over another port. If a connector is **half-mated**, lift it clear with the **right gripper** and mate it again before going on.

Expected state: all six connectors are fully mated, the lead tray is empty, the nut dish is empty, the panel is still square inside the assembly spot, and the left arm is still at home.

Step 2: Tug-test the six mated connectors

Goal: every one of the six connectors is pulled once and holds.

- Tug-test **port 1 first, then 2, 3, 4, 5, 6**. Test every connector, every episode.
- With the **right gripper**, pinch the **lead** just behind its connector — not the connector body, not the pins.
- With the **right gripper**, pull the lead **straight up**, away from the panel. Keep the pull light and steady, for a count of one. The pull stays light enough that the panel does not lift with it.
- Pull straight up only. The **right gripper** never jerks a lead, never pulls it sideways, and never twists it.
- Open the **right gripper**, let go, and watch the connector. It must not have moved.
- If a connector **backs off**, lift it clear with the **right gripper**, mate it again the way its substep in Step 1 says, and tug-test that same connector again. For port 6, close the thumb nut again in quarter turns instead, then tug-test it again.
- If the same connector backs off after two re-mates, leave that port as it is, go on to Step 3, and log the panel for a station check.

Check: all six connectors were pulled, and none of them moved. If any connector was not pulled, pull it with the **right gripper** before going on to Step 3.

Expected state: all six connectors are fully mated and hold, every lead lies clear of the other ports, and the panel is still square inside the assembly spot.

Step 3: End the episode

Goal: recording ends with all six connectors mated the right way round and tug-tested.

1. Confirm the end state:
 - all six ports hold the connector from the tray slot with the same number;
 - the RJ45 latch tab, the USB symbol, and the header key ridge all face the **front edge**;
 - the barrel plug stands in line with its jack with no gap at the sleeve face;
 - the spade receptacle sits square on its tab with the collar at the base;
 - the ring terminal lies flat and the thumb nut is closed;
 - every lead lies clear of the other ports;
 - the lead tray and the nut dish are empty and both are still on their spots;
 - the connector panel is still **square** inside the assembly spot.
2. Return the right arm home with its gripper open. The left arm is already home. Homing is the last thing the arm does.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

After each episode

1. Open the thumb nut counter-clockwise until it runs off the ring stud, and lay it back in the nut dish.
2. Unmate the six connectors, port 6 back to port 1. Press the RJ45 latch tab down before lifting that plug. Lift each connector straight up, never sideways.
3. Lay each lead back in the tray slot with its own number.
4. Blow or brush dust out of the six ports and off the panel.
5. Set the connector panel **square** inside the assembly spot, flat against the table, and stand the lead tray in the start zone for the next episode's config — front-center (Config M), front-right (Config R1), or back-right (Config R2) — leaving the other two zones bare.
6. Run both Setup checklists before the next episode.

At the end of the session

1. Inspect the parts. Replace a lead whose connector has a bent pin, a bent or splayed spade receptacle, a broken RJ45 latch tab, a cracked shell, or a lead pulled loose from its connector. Replace the panel if a port no longer holds a connector, the header shroud notch is broken, the spade tab is bent, the ring stud thread is damaged, or the non-slip underside no longer grips the table. Replace the thumb nut if it no longer runs down the stud by hand.

2. Leave the panel square inside the assembly spot with all six ports empty, all six leads in their own tray slots, and the thumb nut in the nut dish.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config R1 (lead tray at the front-right). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the lead tray is not in the start zone for the config; the right gripper reaches across the table's center line for a lead, or the left arm reaches for one; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the lead tray starts at the front-center in Config M, the front-right in Config R1, or the back-right in Config R2; the right gripper picks every lead in every config and never leans across the center line; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the lead tray is before the first reach, then follow that config's IF line through Step 1.

Violation: Parked arm moved

- **Visible cue:** the left arm leaves home, its gripper closes, or it reaches toward, holds, or steadies the panel, a lead, or the thumb nut at any point in the episode.

- **SOP rule broken:** Steps 1 to 3, this is a single-arm task. The left arm stays at home with its gripper open from the start of recording to the end.
- **Coaching note:** the fixtures do the holding. If a step feels like it needs a second gripper, report the panel or the tray for a station check instead of using the left arm.

Violation: Connector panel gripped or moved

- **Visible cue:** the right gripper grips, lifts, turns, or drags the connector panel to steady it or to bring a port closer, or the panel ends the episode outside the assembly spot or out of square with the front edge.
- **SOP rule broken:** Steps 1–2 (the connector panel stays flat and square inside the assembly spot for the whole episode and is never gripped or lifted).
- **Coaching note:** bring the lead to the panel; never move the panel to the lead. Its weight holds it.

Violation: Connectors mated out of order, or a port left empty

- **Visible cue:** the right gripper mates the connectors in some order other than port 1, 2, 3, 4, 5, 6, or the episode moves on to the tug tests or the ending with a port still empty.
- **SOP rule broken:** Step 1 (work port 1 first, then 2, 3, 4, 5, 6, one lead at a time, and mate all six).
- **Coaching note:** left to right, one at a time. Say the port number before you pick the lead.

Violation: Lead taken to the wrong port

- **Visible cue:** the right gripper carries a lead to a port whose number does not match its tray slot, or tries a connector in a port it does not fit.
- **SOP rule broken:** Step 1 (the lead in tray slot N goes in port N).
- **Coaching note:** slot number and port number are the same number. Read the number before you lift.

Violation: Lead mishandled on the way to the port

- **Visible cue:** the right gripper pinches a connector's pins, blade, or metal shell, carries two leads at once, or sets a lead down on the panel or the table on the way from the tray.
- **SOP rule broken:** Step 1 (carry each lead in one go straight from the tray to its port, held across the body of the connector, never on the pins, the blade, or the shell).
- **Coaching note:** one lead, hold the body, straight to the port.

Violation: Connector forced or levered into its port

- **Visible cue:** the right gripper pushes a connector in leaning or from the side, rocks or levers it sideways to make it go, or keeps forcing one that has stopped, instead of lifting it clear and lining it up again.

- **SOP rule broken:** Step 1 (every connector goes in straight down; if it will not go in, lift it clear, turn it the right way round, and push it straight down again).
- **Coaching note:** straight down, no levering. Sideways force is what slides the panel. If it fights you, it is the wrong way round.

Violation: RJ45 mated wrong

- **Visible cue:** the right gripper pushes the RJ45 plug in with the latch tab facing away from the front edge, or lets go with no click and the plug standing proud of the jack.
- **SOP rule broken:** Step 1.1 (turn the plug so the latch tab faces the front edge, and push straight down until it clicks).
- **Coaching note:** tab to the front, push until you hear and see the click.

Violation: USB plug mated wrong

- **Visible cue:** the right gripper pushes the USB plug in with the symbol facing away from the front edge, or lets go with the shell standing out and a gap at the face of the port.
- **SOP rule broken:** Step 1.2 (turn the plug so the USB symbol faces the front edge, and push straight down until the shell is flush).
- **Coaching note:** symbol to the front, in until the shell is flush. A USB that stops is the wrong way round.

Violation: Barrel plug not seated square

- **Visible cue:** the right gripper leaves the barrel plug cocked in its jack, or lets go with a gap showing at the sleeve face.
- **SOP rule broken:** Step 1.3 (stand the barrel plug in line with the jack and push straight down until the sleeve face touches the face of the jack).
- **Coaching note:** it is round, so nothing to line up but the axis. Stand it up straight and press it home.

Violation: Header plug mated wrong

- **Visible cue:** the right gripper turns the header key ridge away from the front edge, pushes the housing against the shroud wall instead of into the notch, or lets go with the housing standing high on the shroud.
- **SOP rule broken:** Step 1.4 (turn the key ridge to the front edge, line it up with the notch, and push straight down until the housing is bottomed).
- **Coaching note:** ridge to the front, into the notch, down until the rim sits on the shroud.

Violation: Spade receptacle mated wrong

- **Visible cue:** the right gripper brings the spade receptacle down with its flats crossed to the tab, catches the edge of the tab, bends the tab over, or lets go with the collar standing off the base of the tab.

- **SOP rule broken:** Step 1.5 (line the receptacle's flats up with the flat faces of the tab and push straight down until the collar meets the base of the tab).
- **Coaching note:** flats along flats. If the tab bends, you were crossed.

Violation: Ring terminal not laid flat

- **Visible cue:** the ring terminal sits tilted or lifted off the panel around the stud, or its lead runs away from the front edge, and the thumb nut goes on anyway.
- **SOP rule broken:** Step 1.6 (press the ring terminal down flat on the panel with its lead running toward the front edge before starting the nut).
- **Coaching note:** flat and pointing front before the nut goes on.

Violation: Thumb nut left open or turned wrong

- **Visible cue:** the episode moves on with the thumb nut standing part way up the stud or still in the nut dish, or the right gripper turns the nut counter-clockwise, turns more than a quarter turn in one grip, or keeps forcing it after it has stopped.
- **SOP rule broken:** Step 1.6 (turn the nut clockwise, a quarter turn per grip, until it stops turning, and stop once it stops).
- **Coaching note:** clockwise, small bites, let go and re-grip. Stop when it stops.

Violation: Half-mated connector left as is

- **Visible cue:** a connector stops short of its tell — no click, a gap at the face, a shell standing proud, a housing standing high — and the right gripper lets go and moves on to the next port.
- **SOP rule broken:** Step 1 (if a connector is half-mated, lift it clear and mate it again before going on).
- **Coaching note:** let go and look before you move. A half-mated connector never fixes itself.

Violation: Lead left lying over another port

- **Visible cue:** a mated lead is left lying across a port that has not been mated yet, and the next connector is brought down through it or the lead is pushed aside mid-mate.
- **SOP rule broken:** Step 1 (after each connector, lay its lead down lying clear of the other ports).
- **Coaching note:** drop the lead away from the row before you pick up the next one.

Violation: Mated connector disturbed

- **Visible cue:** while mating or tug-testing one port, the right gripper knocks, drags, or pulls a neighbouring lead, and that connector lifts, backs off, or comes out.
- **SOP rule broken:** Steps 1–2 (a connector that is fully mated stays fully mated for the rest of the episode).

- **Coaching note:** come in from above and keep the carry short. Look at what else your gripper is touching.

Violation: Tug test skipped

- **Visible cue:** the right gripper pulls fewer than six leads, or no lead at all, before the episode moves on to the ending.
- **SOP rule broken:** Step 2 (tug-test port 1 through port 6; test every connector, every episode).
- **Coaching note:** six connectors, six pulls. Count them out as you go.

Violation: Tug test done the wrong way

- **Visible cue:** the right gripper jerks a lead, pulls it sideways, twists it, pinches the connector body or its pins instead of the lead, or pulls hard enough that the panel lifts off the table.
- **SOP rule broken:** Step 2 (pinch the lead just behind its connector and pull straight up, light and steady, for a count of one, light enough that the panel does not lift).
- **Coaching note:** light, steady, straight up, on the lead. If the panel lifts, you are pulling far too hard.

Violation: Failed tug test not fixed and re-tested

- **Visible cue:** a connector lifts, slides, or comes out during its tug test and the episode moves on; or the right gripper re-mates it and never pulls that connector again.
- **SOP rule broken:** Step 2 (on a connector that backs off, mate it again the way its Step 1 substep says, then tug-test that same connector again).
- **Coaching note:** every re-mate earns another pull on that same connector.

Violation: Work done out of order

- **Visible cue:** the phases run out of sequence — the right gripper tug-tests a connector before all six are mated, or goes back to mate a port after the tug tests have started.
- **SOP rule broken:** Steps 1–2 (mate all six, then tug-test all six, in that order).
- **Coaching note:** finish the phase before you start the next one. Say the phase name before you move.

Violation: End-state check skipped

- **Visible cue:** recording stops, or the right arm goes home, with no look across the six ports, and the episode ends with a wrong-way-round connector, a half-mated connector, or an open thumb nut still in view.
- **SOP rule broken:** Step 3 (confirm the end state, item by item, before the right arm goes home).
- **Coaching note:** run the list out loud across the row before you home the arm.

Violation: Lead, nut, or tray knocked off its spot

- **Visible cue:** the right gripper drops a lead on the panel, the table, or the floor, drops the thumb nut, or tips, pushes, or spills the lead tray or the nut dish off its spot.
- **SOP rule broken:** Steps 1–2 (keep the leads, the thumb nut, the lead tray, and the nut dish on their spots through the whole episode).
- **Coaching note:** work slower and lower over the table, and keep each carry short.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the six ports are confirmed, the right arm is away from home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 3 (confirm the end state, return the right arm home with its gripper open as its final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arm does.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- A lead arrives with a bent pin, a broken RJ45 latch tab, a cracked shell, or the connector already loose on the lead, so it cannot mate.
- The RJ45 jack's latch slot is worn out before the episode, so a correctly pushed plug never clicks.
- The USB port's plastic tongue is broken before the episode, so a correctly turned plug will not stay in.
- The header shroud's notch is broken before the episode, so a correctly turned key ridge does not line up.
- The spade tab arrives bent, so a correctly lined-up receptacle cannot slide on.
- The ring stud thread is stripped before the episode, so the thumb nut never closes.
- The panel's non-slip underside has failed before the episode, so the panel slides under a straight-down push that was done correctly.

Annotation subtasks (from SOP)

1. Pick a lead from the tray slot with the port's number
2. Turn the connector so its keyed feature faces the front edge
3. Push the connector straight down into its port until it is fully mated

4. Lay the mated lead down clear of the other ports
5. Repeat the pick, turn, and push for all six ports
6. Drop the ring terminal over the stud and press it flat
7. Pick the thumb nut from the nut dish and start it on the stud
8. Turn the thumb nut down in quarter turns until it stops
9. Tug-test each connector, port 1 through 6
10. Re-mate and re-test any connector that backs off